

Template-repo bootstrap

Course labs often start from a template repository, not a blank folder

Clone, map, status

- Clone the template URL into a project folder, change into it, and list the top level
- Expect directories like rtl, tb, scripts
- Run status right away
- Read gitignore and env example files so you know what not to commit
- Map the layout before you write new code

Browser lab

Digital Design and Verification Platform

HomeTools

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INTERACTIVE TOOL

Template-repo bootstrap

Clone a chip **template**, inspect the layout, then run your first `git status` — clean tree, origin, and main.

Starter example: Empty home. Clone chip-template into chip-lab (or use-template), cd in, inspect layout, then `git status`.

Load starter example

Challenges 0 / 22 cleared

Quiz: writable copy: GitHub button for your own writable copy? Answer: use this template

Answer

Show hintCheckNextIdle

Quiz: writable copyQuiz: cloneQuiz: first checkQuiz: remote nameClone or templateEnter repoFirst status

See originSee branchTree layoutRead READMERead MakefileQuiz: RTL dirQuiz: TB dirUse template path

Clone pathls topRead gitignoreQuiz: clean phraseMust cd firstQuiz: default branchFull bootstrap

Core ideas

Template → your copy

Use this template (writable) or `git clone` the template URL.

First status

After clone: clean tree on main, origin points at the remote.

Lab terminal

Layout & file

Template clone lab starter

Real Git practice

```
wsl - module19-template-clone/examples/template

# Track A - real Unix
# real shell session (Track A)

# practice sandbox: /tmp/git-tc-g3stek

$ git clone ../chip-template chip-lab

$ cd chip-lab

/tmp/git-tc-g3stek/chip-lab

$ ls -la
total 36
drwxr-xr-x 7 yongfu yongfu 4096 Jul 20 14:51 .
drwx----- 4 yongfu yongfu 4096 Jul 20 14:51 ..
drwxr-xr-x 8 yongfu yongfu 4096 Jul 20 14:51 .git
-rw-r--r-- 1 yongfu yongfu 7 Jul 20 14:51 .gitignore
-rw-r--r-- 1 yongfu yongfu 44 Jul 20 14:51 README.md
drwxr-xr-x 2 yongfu yongfu 4096 Jul 20 14:51 docs
drwxr-xr-x 2 yongfu yongfu 4096 Jul 20 14:51 rtl
drwxr-xr-x 2 yongfu yongfu 4096 Jul 20 14:51 scripts
drwxr-xr-x 2 yongfu yongfu 4096 Jul 20 14:51 tb

$ ls rtl tb scripts docs
docs:
README.md

rtl:
alu.v

scripts:
check_ready.sh

tb:
tb_alu.v

$ cat README.md
# Chip lab template

Quick start: make help

$ git status
## main...origin/main

$ git remote -v
origin /tmp/git-tc-g3stek/chip-template (fetch)
origin /tmp/git-tc-g3stek/chip-template (push)

$ git branch
* main
```

Real shell — clone template and inspect

Real Git practice — try these

```
# (setup) chip-template repo with rtl/, tb/, scripts/, docs/

# git clone <template-url> chip-lab — copy template into your
workspace
git clone ../chip-template chip-lab

# cd chip-lab — work inside the new project
cd chip-lab

# ls -la — see top-level files and folders
ls -la

# ls rtl tb scripts docs — map source, tests, and tooling
ls rtl tb scripts docs

# cat README.md — read quick-start notes
```

Pitfalls to watch

- Clone into a fresh folder name, you do not want to nest a repo inside another by accident
- If status is not clean immediately after clone, find out why before you start editing
- Do not commit env files with secrets, copy from env example instead
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter, bootstrap chip-lab, and run your first status
- On real Git, clone a template tree and map rtl, tb, and scripts
- When you are ready